

Ren Pang

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EDUCATION

Ph.D.	<i>Information Sciences and Technology</i>	Pennsylvania State University	2019 – 2023
B.Sc.	<i>Mathematics</i>	Nankai University	2014 – 2018

WORK EXPERIENCE

Applied Scientist, <i>Amazon</i>	2024 – Present
Develop Guardrails for Amazon Bedrock, the safeguards customized to customers' application requirements and responsible AI policies.	

INTERN EXPERIENCE

Applied Scientist (Intern), <i>Amazon</i>	2023 Summer
Explore the vulnerabilities of LLMs to jail-breaking attacks, where Reinforcement Learning from Human Feedback (RLHF) is considered to enhance the attack and defense efficiency. During project development, I submitted several bug fixes and new features to Transformers, Peft and Trl libraries.	
Machine Learning Engineer (Intern), <i>Meta</i>	2022 Summer
Pages and Groups Integrity: Introduce new classification model for malicious page detection. It mitigates the impact of incorrect label annotation, and provides interpretable classification outputs for better user experience.	
TorchVision: Provide the official TorchVision implementation of SwinTransformerV2.	

PUBLICATIONS

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1. Watch the Watcher! Backdoor Attacks on Security-Enhancing Diffusion Models, C. Li, **R. Pang**, B. Cao, J. Chen, F. Ma, S. Ji, and T. Wang, Proceedings of the *USENIX Security Symposium (USENIX)*, 2025.
 2. On the Difficulty of Defending Contrastive Learning against Backdoor Attacks, C. Li, **R. Pang**, B. Cao, J. Chen, S. Ji, and T. Wang, Proceedings of the *USENIX Security Symposium (USENIX)*, 2024.
 3. Defending Pre-trained Language Models as Few-shot Learners Against Backdoor Attacks, Z. Xi, T. Du, C. Li, **R. Pang**, S. Ji, J. Chen, F. Ma, and T. Wang, Proceedings of *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
 4. An Embarrassingly Simple Backdoor Attack against Self-supervised Learning, C. Li, **R. Pang**, Z. Xi, T. Du, S. Ji, Y. Yao, and T. Wang, Proceedings of the *International Conference on Computer Vision (ICCV)*, 2023.
 5. On the Security Risks of Knowledge Graph Reasoning, Z. Xi, T. Du, C. Li, **R. Pang**, S. Ji, X. Luo, X. Xiao, F. Ma, and T. Wang, Proceedings of the *USENIX Security Symposium (USENIX)*, 2023.
 6. The Dark Side of AutoML: Towards Architectural Backdoor Search, **R. Pang**, C. Li, Z. Xi, S. Ji, and T. Wang, Proceedings of the *International Conference on Learning Representations (ICLR)*, 2023.
 7. TrojanZoo: Towards Unified, Holistic, and Practical Evaluation of Neural Backdoors, **R. Pang**, Z. Zhang, X. Gao, Z. Xi, S. Ji, P. Cheng, and T. Wang, Proceedings of the *IEEE European Symposium on Security and Privacy (EuroS&P)*, 2022.
 8. On the Security Risks of AutoML, **R. Pang**, Z. Xi, S. Ji, X. Luo, and T. Wang, Proceedings of the *USENIX Security Symposium (USENIX)*, 2022.

9. Graph Backdoor,
Z. Xi, **R. Pang**, S. Ji, and T. Wang,
Proceedings of *the USENIX Security Symposium (USENIX)*, 2021.
10. i-Algebra: Towards Interactive Interpretability of Deep Neural Networks,
X. Zhang, **R. Pang**, S. Ji, F. Ma, and T. Wang,
Proceedings of *the AAAI Conference on Artificial Intelligence (AAAI)*, 2021.
11. AdvMind: Inferring Adversary Intent of Black-Box Attacks,
R. Pang, X. Zhang, S. Ji, X. Luo, and T. Wang,
Proceedings of *the ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2020.
12. A Tale of Evil Twins: Adversarial Inputs versus Poisoned Models,
R. Pang, H. Shen, X. Zhang, S. Ji, Y. Vorobeychik, X. Luo, A. Liu, and T. Wang,
Proceedings of *the ACM Conference on Computer and Communications Security (CCS)*, 2020.

OPEN-SOURCE CONTRIBUTION

1. **TrojanZoo** (*owner*)  <https://github.com/ain-soph/trojanzoo>
(70,000 Lines) Offer a universal, flexible PyTorch platform to conduct security analysis of attacks and defenses on deep neural network models.
2. **AlpsPlot** (*owner*)  <https://github.com/ain-soph/alpsplot>
(14,000 Lines) Offer a high-level python library to plot academic figures based on Matplotlib.
3. **TorchVision.SwinTransformerV2**  <https://github.com/pytorch/vision/pull/6246>
(400 Lines) Provide TorchVision official implementation of SwinTransformerV2.
4. **Matplotlib.Text**  <https://github.com/matplotlib/matplotlib/pull/20101>
Fix Text class bug when font argument is provided without math_fontfamily.